

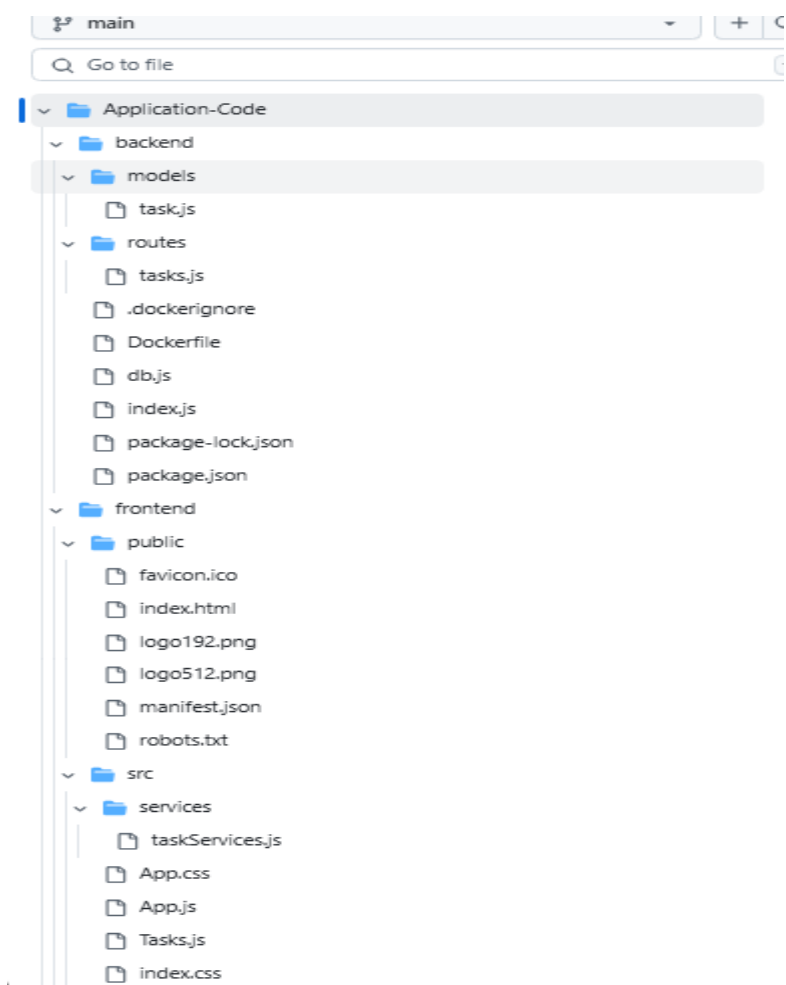
Capstone Project: Product APP

Git repository

 root	updating	7b6d617 · 4 days ago	 21 Commits
 Application-Code	Added Project Enhancement		last year
 Jenkins-Pipeline-Code	Added Project Enhancement		last year
 Jenkins-Server-TF	Added Project Enhancement		last year
 Kubernetes-Manifests-file	updating		4 days ago
 assets	Added Project Enhancement		last year
 README.md	Update README.md		6 months ago
 buildspec.yml	edited buildspec.yml		5 days ago
 iam_policy.json	updating		4 days ago

 index.css	
 index.js	
 .dockerignore	
 Dockerfile	
 package-lock.json	
 package.json	
>  Jenkins-Pipeline-Code	
>  Jenkins-Server-TF	
✓  Kubernetes-Manifests-file	
✓  Backend	
 deployment.yaml	
 service.yaml	
✓  Database	
 deployment.yaml	
 pv.yaml	
 pvc.yaml	
 secrets.yaml	
 service.yaml	
✓  Frontend	
 deployment.yaml	
 iam_policy.json	
 service.yaml	
 ingress.yaml	
✓  assets	
 Three-Tier.gif	

- package-lock.json
- package.json
- > Jenkins-Pipeline-Code
- > Jenkins-Server-TF
- ▼ Kubernetes-Manifests-file
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 - deployment.yaml
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 - ▼ Frontend
 - deployment.yaml
 - iam_policy.json
 - service.yaml
 - ingress.yaml
 - ▼ assets
 - Three-Tier.gif
 - README.md
 - buildspec.yml
 - iam_policy.json



```
root@ip-172-31-93-34:~/react-product-app/k8s/frontend# kubectl get deploy -n project
```

NAME	READY	UP-TO-DATE	AVAILABLE	AGE
frontdeploy	1/1	1	1	29s

Buildspec.yml

Code Blame 91 lines (78 loc) · 4.12 KB

Raw Download Edit View

```
1 version: 0.2
2
3 env:
4   variables:
5     ACCOUNT_ID: 026098545815
6     REGION: us-east-1
7     FRONTEND_REPO: projectfront
8     BACKEND_REPO: projectback
9     FRONTEND_DIR: frontend
10    BACKEND_DIR: backend
11    KBS_MANIFEST_DIR: k8s
12
13 phases:
14   install:
15     runtime-versions:
16       nodejs: 18
17       java: corretto17
18     commands:
19       - echo Installing utilities...
20       - yum install -y jq
21       - echo Installing Sonar Scanner...
22       - curl -sLo sonar-scanner-cli.zip https://binaries.sonarsource.com/Distribution/sonar-scanner-cli/sonar-scanner-cli-4.8.0.2856-linux.zip
23       - unzip sonar-scanner-cli.zip
24       # Add sonar-scanner to PATH environment variable for subsequent phases by exporting and saving in environment
25       - export PATH=$PATH:$PWD/sonar-scanner-4.8.0.2856-linux/bin
26       - echo "export PATH=$PATH:$PWD/sonar-scanner-4.8.0.2856-linux/bin" >> $HOME/.bashrc
27
```

Code Blame 91 lines (78 loc) · 4.12 KB

Raw Download Edit View

```
28 pre_build:
29   commands:
30     - echo Setting dynamic image tag...
31     - export IMAGE_TAG=v1.${CODEBUILD_BUILD_NUMBER}
32     - echo IMAGE_TAG=$IMAGE_TAG
33     - echo Logging in to Amazon ECR...
34     - aws ecr get-login-password --region $REGION | docker login --username AWS --password-stdin ${ACCOUNT_ID}.dkr.ecr.${REGION}.amazonaws.com
35     - echo Installing frontend dependencies...
36     - cd $FRONTEND_DIR && npm install && cd -
37     - echo Building backend with Maven...
38     - cd $BACKEND_DIR && mvn clean package -DskipTests && cd -
39
40 build:
41   commands:
42     - echo Adding sonar-scanner to PATH...
43     - export PATH=$PATH:$PWD/sonar-scanner-4.8.0.2856-linux/bin
44     - echo Running SonarQube scanner...
45     - sonar-scanner -Dsonar.projectKey=hello -Dsonar.sources=backend/src -Dsonar.java.binaries=backend/target/classes -Dsonar.host.url=http://34.228.190.63 -Dsonar.login=sqp_4fa73fba54
46
47     - echo Building Docker images...
48     - docker build -t $FRONTEND_REPO:$IMAGE_TAG $FRONTEND_DIR || { echo "Frontend Docker build failed"; exit 1; }
49     - docker tag $FRONTEND_REPO:$IMAGE_TAG ${ACCOUNT_ID}.dkr.ecr.${REGION}.amazonaws.com/$FRONTEND_REPO:$IMAGE_TAG
50     - docker build -t $BACKEND_REPO:$IMAGE_TAG $BACKEND_DIR || { echo "Backend Docker build failed"; exit 1; }
51     - docker tag $BACKEND_REPO:$IMAGE_TAG ${ACCOUNT_ID}.dkr.ecr.${REGION}.amazonaws.com/$BACKEND_REPO:$IMAGE_TAG
52     - docker images
53
54 post_build:
55   commands:
56     - echo Pushing Docker images to ECR...
57     - docker push ${ACCOUNT_ID}.dkr.ecr.${REGION}.amazonaws.com/$FRONTEND_REPO:$IMAGE_TAG
58     - docker push ${ACCOUNT_ID}.dkr.ecr.${REGION}.amazonaws.com/$BACKEND_REPO:$IMAGE_TAG
59
```

```
react-product-app / buildspec.yml
Code Blame 91 lines (78 loc) · 4.12 KB
- echo Pushing Docker images to ECR...
- docker push ${ACCOUNT_ID}.dkr.ecr.${REGION}.amazonaws.com/$FRONTEND_REPO:$IMAGE_TAG
- docker push ${ACCOUNT_ID}.dkr.ecr.${REGION}.amazonaws.com/$BACKEND_REPO:$IMAGE_TAG
- echo Preparing Kubernetes manifest output directory...
- mkdir -p k8s-out
- touch k8s-out/.keep # Prevent "directory not found" if no files copied
- sed "s|IMAGE_PLACEHOLDER|${ACCOUNT_ID}.dkr.ecr.${REGION}.amazonaws.com/$FRONTEND_REPO:$IMAGE_TAG|g" $K8S_MANIFEST_DIR/frontend/deployment.yaml > k8s-out/frontend-deployment.yaml
- cp $K8S_MANIFEST_DIR/frontend/service.yaml k8s-out/frontend-service.yaml
- sed "s|IMAGE_PLACEHOLDER|${ACCOUNT_ID}.dkr.ecr.${REGION}.amazonaws.com/$BACKEND_REPO:$IMAGE_TAG|g" $K8S_MANIFEST_DIR/backend/deployment.yaml > k8s-out/backend-deployment.yaml
- cp $K8S_MANIFEST_DIR/backend/service.yaml k8s-out/backend-service.yaml
- cp $K8S_MANIFEST_DIR/namespace.yaml k8s-out/namespace.yaml || true
- cp $K8S_MANIFEST_DIR/database/pv.yaml k8s-out/database-pv.yaml || true
- cp $K8S_MANIFEST_DIR/database/pvc.yaml k8s-out/database-pvc.yaml || true
- cp $K8S_MANIFEST_DIR/database/app-config.yaml k8s-out/database-app-config.yaml || true
- cp $K8S_MANIFEST_DIR/database/secrets.yaml k8s-out/database-secrets.yaml || true
- cp $K8S_MANIFEST_DIR/database/deployment.yaml k8s-out/database-deployment.yaml || true
- cp $K8S_MANIFEST_DIR/database/service.yaml k8s-out/database-service.yaml || true
- cp $K8S_MANIFEST_DIR/ingress.yaml k8s-out/ingress.yaml || true
- echo "Final list of generated Kubernetes YAML files:"
- find k8s-out/
artifacts:
  base-directory: k8s-out
  discard-paths: yes
  files:
    - '**/*.yaml'
```

Frontend service

```
GNU nano 7.2 service.yaml
apiVersion: v1
kind: Service
metadata:
  name: react-frontend-service
  namespace: project
spec:
  type: LoadBalancer
  selector:
    app: react-frontend
  ports:
    - port: 80
      targetPort: 80
```

```
root@ip-172-31-93-34:~/react-product-app/k8s/frontend# kubectl expose deployment frontend --type=LoadBalancer --port=80 --target-port=80 --name=frontservice -n project -o yaml > service.yaml
```

```
root@ip-172-31-93-34:~/react-product-app/k8s/database# kubectl apply -f pv.yaml
persistentvolume/mysql-pv created
root@ip-172-31-93-34:~/react-product-app/k8s/database# kubectl apply -f pvc.yaml
persistentvolumeclaim/mysql-volume-claim created
root@ip-172-31-93-34:~/react-product-app/k8s/database# kubectl get pv,pvc -n project
NAME                                     CAPACITY  ACCESS MODES  RECLAIM POLICY  STATUS  CLAIM                                     STORAGECLASS  VOLUMEATTRIBUTESCLASS  REASON  AGE
persistentvolume/mysql-pv              1Gi       RWO           Retain          Bound   project/mysql-volume-claim              <unset>
NAME                                     STATUS  VOLUME  CAPACITY  ACCESS MODES  STORAGECLASS  VOLUMEATTRIBUTESCLASS  AGE
persistentvolumeclaim/mysql-volume-claim Bound   mysql-pv  1Gi       RWO           <unset>          <unset>                  21s
```

Stacks

Stacks (14)

 *Filter by stack name*

Filter status

Active 

 View nested

< **1** >

Stacks

cloud



2025-06-23 19:18:47 UTC+0530



CREATE_COMPLETE

Stacks (14)

Filter by stack name

Filter status

Active

View nested

Stacks

cloud

2025-06-23 19:18:47 UTC+0530

CREATE_COMPLETE

CLOUDFOR

2025-06-23 18:44:55 UTC+0530

DELETE_IN_PROGRESS

eksctl-project-cluster-addon-

iamserviceaccount-kube-system-

Stack info

Events

Resources

Outputs

Parameters

Template

Table view

Timeline view

Events (91)

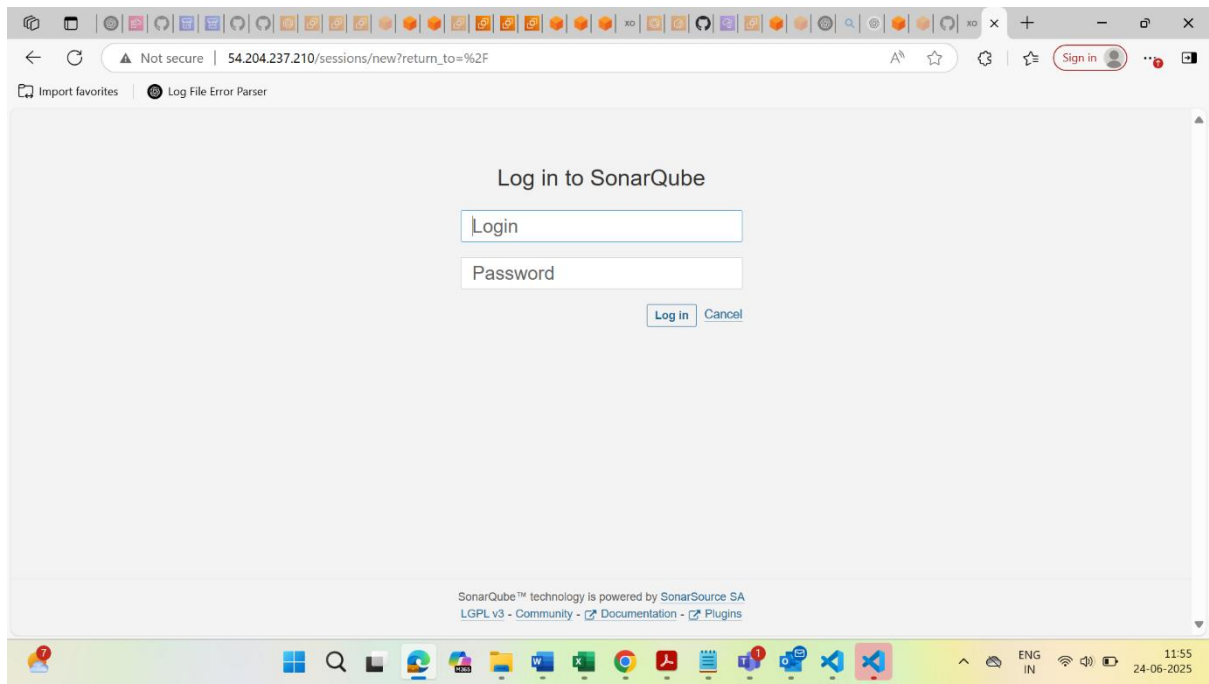
View root cause

Search events

Timestamp	Logical ID	Status	Detailed st
2025-06-23 19:32:01 UTC+0530	cloud	CREATE_COMPLETE	-
2025-06-23 19:31:59 UTC+0530	MySQLDatabase	CREATE_COMPLETE	-
2025-06-23 19:28:04 UTC+0530	MyNodeGroup	CREATE_COMPLETE	-

```
root@ip-10-0-2-197:~/react-product-app/backend# docker tag projectback:latest 026090545815.dkr.ecr.us-east-1.amazonaws.com/projectback:latest
root@ip-10-0-2-197:~/react-product-app/backend# docker push 026090545815.dkr.ecr.us-east-1.amazonaws.com/projectback:latest
The push refers to repository [026090545815.dkr.ecr.us-east-1.amazonaws.com/projectback]
52638d05a81e: Layer already exists
ed1649494802: Layer already exists
b7043f8985d3: Layer already exists
dd93bf259fd4: Layer already exists
latest: digest: sha256:f6ff82dl3470fd0lc7ab237b34896fcb7c49d53fle20d4c1074badal5398895f size: 1160
root@ip-10-0-2-197:~/react-product-app/backend# kubectl delete pod -n project -l app=springboot-backend
kubectl delete pod -n project -l app=springboot-backend
pod "backdeploy-64d676dcd5-hs9c5" deleted
pod "springboot-backend-5dd9d868cd-vckfn" deleted
root@ip-10-0-2-197:~/react-product-app/backend# kubectl rollout restart deployment backdeploy -n project
kubectl rollout restart deployment springboot-backend -n project
deployment.apps/backdeploy restarted
deployment.apps/springboot-backend restarted
root@ip-10-0-2-197:~/react-product-app/backend# kubectl get po -n project
NAME                                READY    STATUS    RESTARTS   AGE
backdeploy-867788ccd8-5bcv4          1/1     Running   0           18s
frontdeploy-588ffc66df-p98fx         1/1     Running   0           8h
springboot-backend-889dc7fdb-nt62s   1/1     Running   0           17s
root@ip-10-0-2-197:~/react-product-app/backend# kubectl get deploy -n project
NAME                                READY    UP-TO-DATE    AVAILABLE    AGE
backdeploy                          1/1      1              1             8h
frontdeploy                         1/1      1              1             8h
springboot-backend                  1/1      1              1             8h
```

Sonar qube



Terraform

```

! CloudFormation.yaml > {} Parameters > {} MyDBSubnetGroupName
1  AWSTemplateFormatVersion: '2010-09-09'
2  Description: StackSet template to create a basic VPC with public and private subnets, NAT, and an RDS MySQL DB
3
4  Parameters:
5    MyVpcCidr:
6      Type: String
7      Description: "CIDR block for the VPC"
8      Default: "10.0.0.0/16"
9      AllowedValues:
10       - "10.0.0.0/16"
11       - "20.0.0.0/18"
12    MyDBSubnetGroupName:
13      Type: String
14      Description: "Name for the DB Subnet Group"
15      Default: "MyDBSubnetGroup"
16    MyDBName:
17      Type: String
18      Description: "Name for the MySQL Database"
19      Default: "MySQLDatabase"
20    MyDBRootUserName:
21      Type: String
22      Description: "Root username for the MySQL Database"
23      Default: "admin"
24    MyDBRootUserPassword:
25      Type: String
26      Description: "Root password for the MySQL Database"
27      Default: "Gitust12"
28    NoEcho: true

```

```

! CloudFormation.yaml > {} Mappings > {} SubnetConfig > {} Vpc20
4  Parameters:
24    MyDBRootUserPassword:
29
30  Mappings:
31    SubnetConfig:
32      Vpc10:
33        PublicSubnetAZ1CIDR: "10.0.0.0/20"
34        PrivateSubnetAZ1CIDR1: "10.0.16.0/20"
35        PrivateSubnetAZ1CIDR2: "10.0.32.0/20"
36        PublicSubnetAZ2CIDR: "10.0.48.0/20"
37        PrivateSubnetAZ2CIDR1: "10.0.64.0/20"
38        PrivateSubnetAZ2CIDR2: "10.0.80.0/20"
39      Vpc20:
40        PublicSubnetAZ1CIDR: "20.0.0.0/22"
41        PrivateSubnetAZ1CIDR1: "20.0.4.0/22"
42        PrivateSubnetAZ1CIDR2: "20.0.8.0/22"
43        PublicSubnetAZ2CIDR: "20.0.12.0/22"
44        PrivateSubnetAZ2CIDR1: "20.0.16.0/22"
45        PrivateSubnetAZ2CIDR2: "20.0.20.0/22"
46
47
48  Conditions:
49    is_cidr_10: !Equals [!Ref MyVpcCidr, "10.0.0.0/16"]
50
51  Resources:
52    MyVpc:
53      Type: AWS::EC2::VPC
54      Properties:
55        CidrBlock: !Ref MyVpcCidr

```

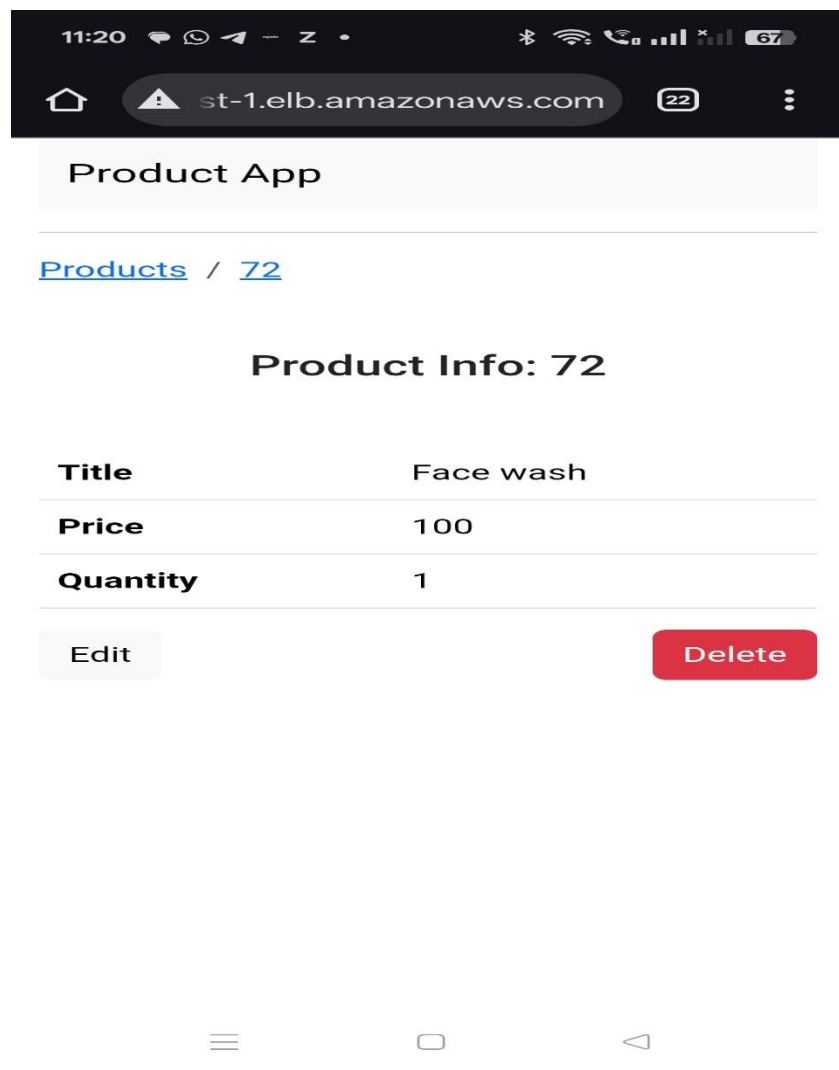


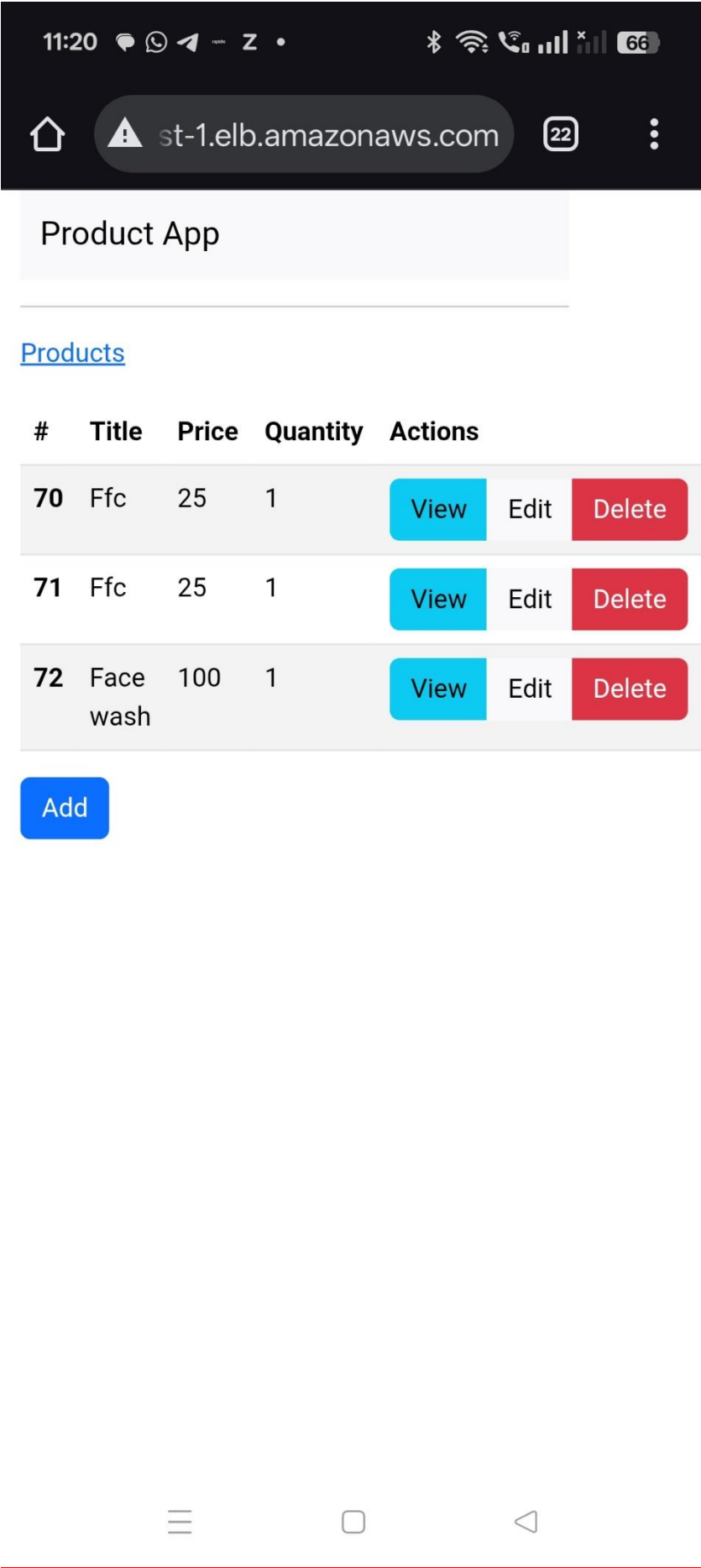
```

51 Resources:
52   MyVpc:
53     Properties:
54       EnableDnsSupport: true
55       EnableDnsHostnames: true
56       Tags:
57         - Key: Name
58           Value: !Sub "${AWS::StackName}-My-VPC"
59
60   MyIGW:
61     Type: AWS::EC2::InternetGateway
62     Properties:
63       Tags:
64         - Key: Name
65           Value: !Sub "${AWS::StackName}-Internet-Gateway"
66
67   MyIGWAttachment:
68     Type: AWS::EC2::VPCGatewayAttachment
69     Properties:
70       VpcId: !Ref MyVpc
71       InternetGatewayId: !Ref MyIGW
72
73   MyPubSub1:
74     Type: AWS::EC2::Subnet
75     Properties:
76       VpcId: !Ref MyVpc
77       AvailabilityZone: !Select [0, !GetAZs '']
78       CidrBlock: !If

```

frontend





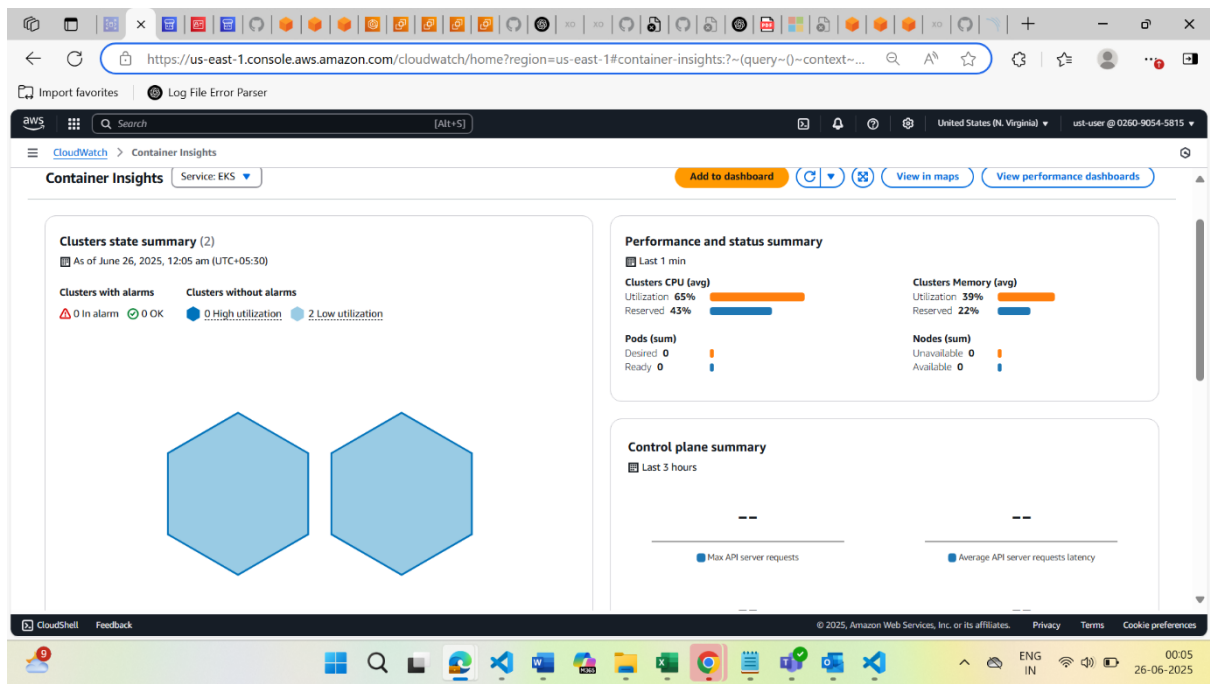
Backend

123	1100	55	Blackberry
124	15	1	Gddh
125	15	1	Gddh
126	1200	99	iPhone
127	1100	55	Blackberry
128	250	1	Food
129	1200	99	iPhone
130	1100	55	Blackberry
131	1200	99	iPhone
132	1100	55	Blackberry
133	1200	99	iPhone
134	1100	55	Blackberry
135	1200	99	iPhone
136	1100	55	Blackberry
137	1	500	Ticket

67 rows in set (0.01 sec)

mysql>

CloudWatch



```
12 # Copy the rest of the application code
13 COPY . .
14
15 # Build the React app
16 RUN npm run build
17
18 # Serve stage
19 FROM public.ecr.aws/nginx/nginx:1.25.1
20
21 # Copy the custom nginx.conf file to the container
22 COPY .docker/nginx.conf /etc/nginx/nginx.conf
23
24 # Copy the built React app from the build stage to the nginx container
25 COPY --from=build /app/build /usr/share/nginx/html
26
27 # Expose port 80
28 EXPOSE 80
29
30 # Start Nginx
31 CMD ["nginx", "-g", "daemon off;"]
```

Database

Databases (5)

Group resources

Modify

Actions

Create database

Filter by databases

< 1 >

DB identifier	Status	Role	Engine	Region ...	Size	Recom
goods	Available	Multi-AZ DB cluster	MySQL Co...	us-east-1	3 instances	1 Inf
goods-instance-1	Available	Writer instance	MySQL Co...	us-east-1c	db.c6gd.medium	
goods-instance-2	Available	Reader instance	MySQL Co...	us-east-1b	db.c6gd.medium	
goods-instance-3	Available	Reader instance	MySQL Co...	us-east-1f	db.c6gd.medium	
mysqlatabase	Available	Instance	MySQL Co...	us-east-1b	db.t3.small	

DB identifier

Status

Role

Engine

Region ...

Size

Recom...

CPU

goods

Available

Multi-AZ ...

MySQL Co...

us-east-1

3 instances

1 Informational

goods-instance-1

Available

Writer ins...

MySQL Co...

us-east-1c

db.c6gd...

15.10%

goods-instance-2

Available

Reader ins...

MySQL Co...

us-east-1b

db.c6gd...

18.85%

goods-instance-3

Available

Reader ins...

MySQL Co...

us-east-1f

db.c6gd...

16.88%

Connectivity & security

Monitoring

Logs & events

Configuration

Zero-ETL integrations

Maintenance & backups

Data migrat

Endpoints (2)

Actions

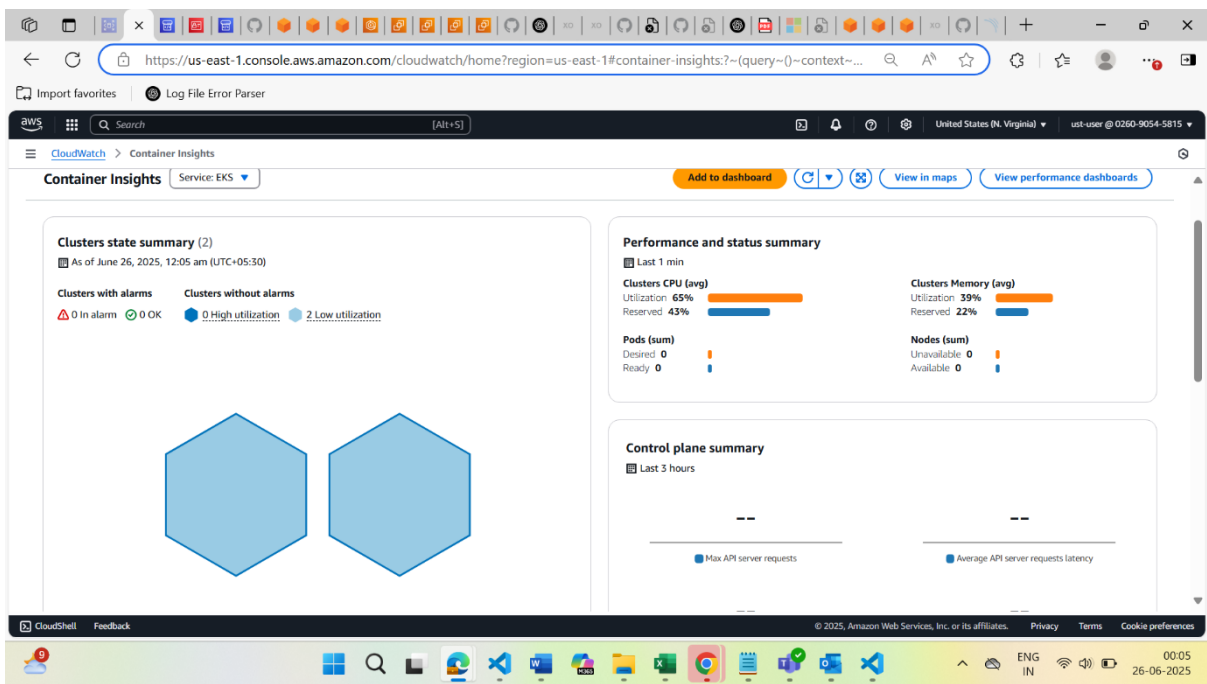
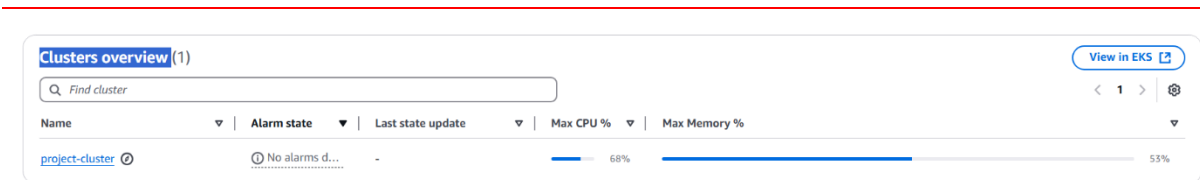
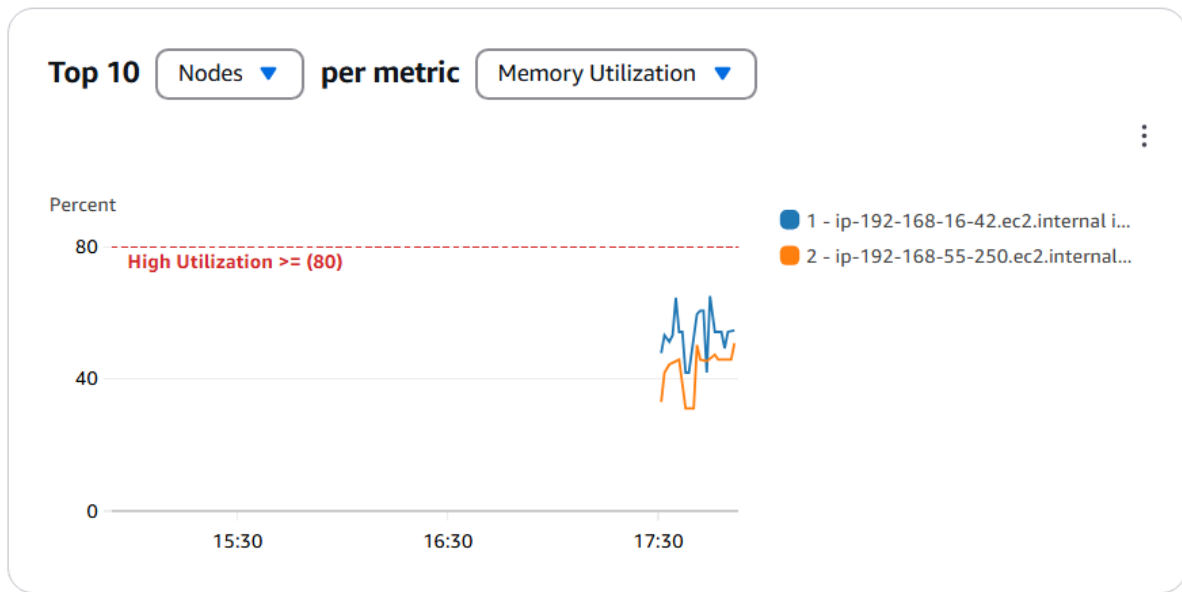
Create custom endpoint

Find resources

< 1 >

Endpoint name	Status	Type	Port
goods.cluster-cf028ok825zn.us-east-1.rds.amazonaws.com	Available	Writer	3306
goods.cluster-ro-cf028ok825zn.us-east-1.rds.amazonaws.com	Available	Reader	3306

cloudwatch



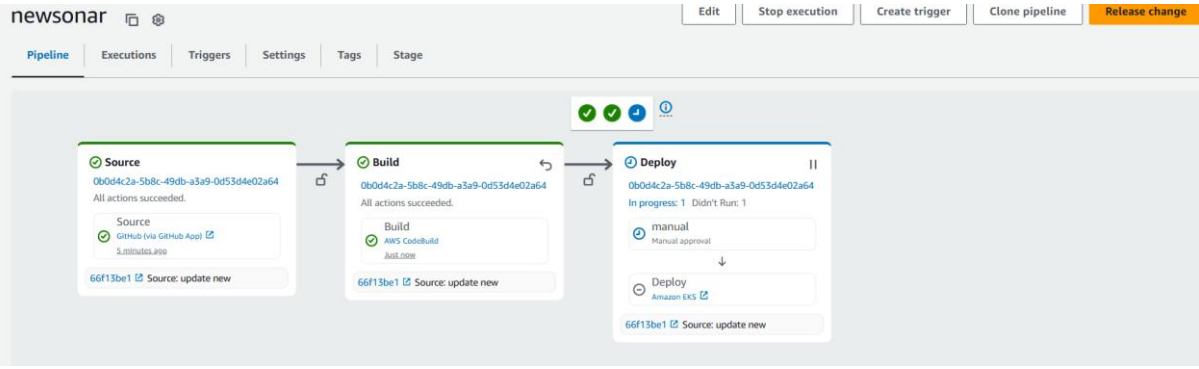
Resources (22)

[View logs](#)
[View dashboard](#)

< 1 2 3 > ⚙

	Name	Type	Alarm state	Max CPU (%)	Max Memory (%)
☐	amazon-cloudwatch	EKS Namespace	① No alarms detected	43.6%	4.2%
☐	aws-load-balancer-controller	EKS Workload	① No alarms detected	0.1%	0.7%
☐	aws-load-balancer-webhook-service	EKS Service	① No alarms detected	0.1%	0.7%
☐	aws-node	EKS Workload	① No alarms detected	0.1%	1.8%
☐	backdeploy	EKS Workload	① No alarms detected	<0.1%	6.8%
☐	backend-service	EKS Service	① No alarms detected	<0.1%	6.8%
☐	cloudwatch-agent	EKS Workload	① No alarms detected	0.2%	1.1%
☐	coredns	EKS Workload	① No alarms detected	<0.1%	0.4%
☐	default	EKS Namespace	① No alarms detected	0.6%	<0.1%
☐	fluentd-cloudwatch	EKS Workload	① No alarms detected	43.6%	4.2%

Codepipeline with Manual approval



Review

Action name: manual Status: Waiting for approval

Details Revisions

Trigger

StartPipelineExecution - [ust-user](#)

Comments about this action

-

URL for review

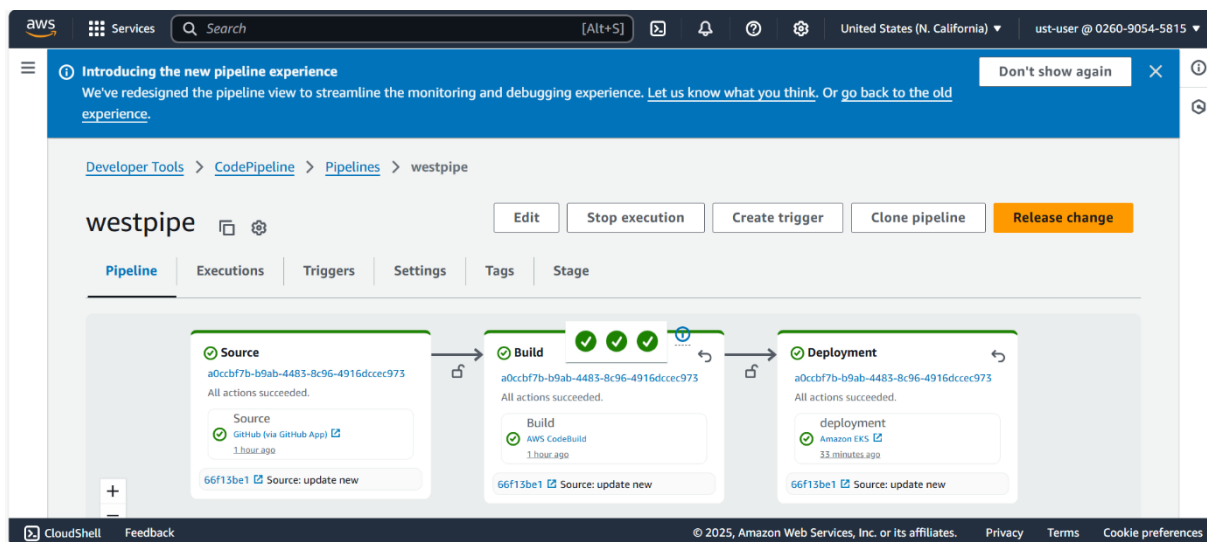
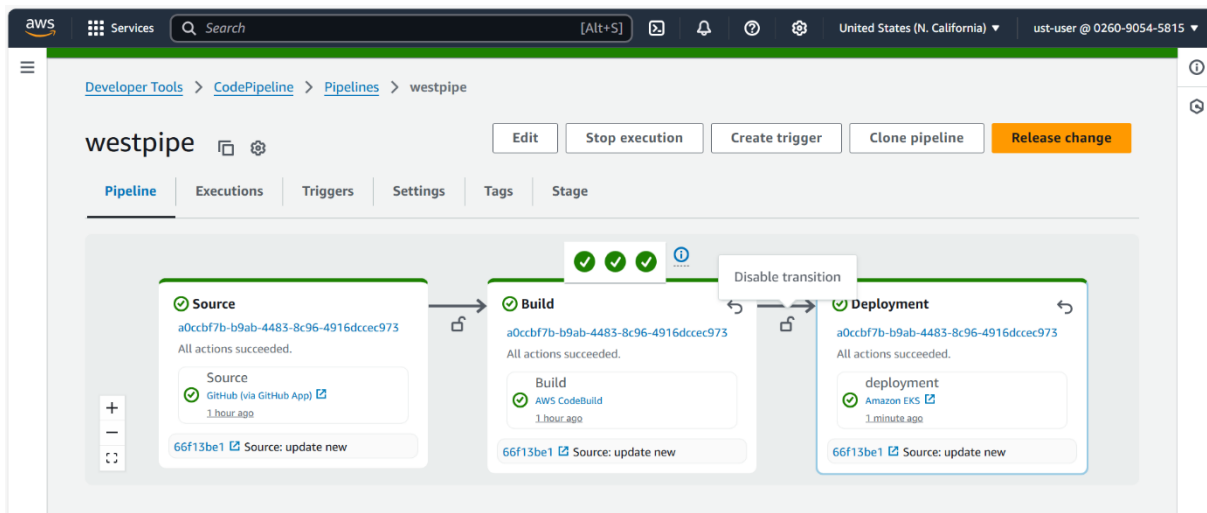
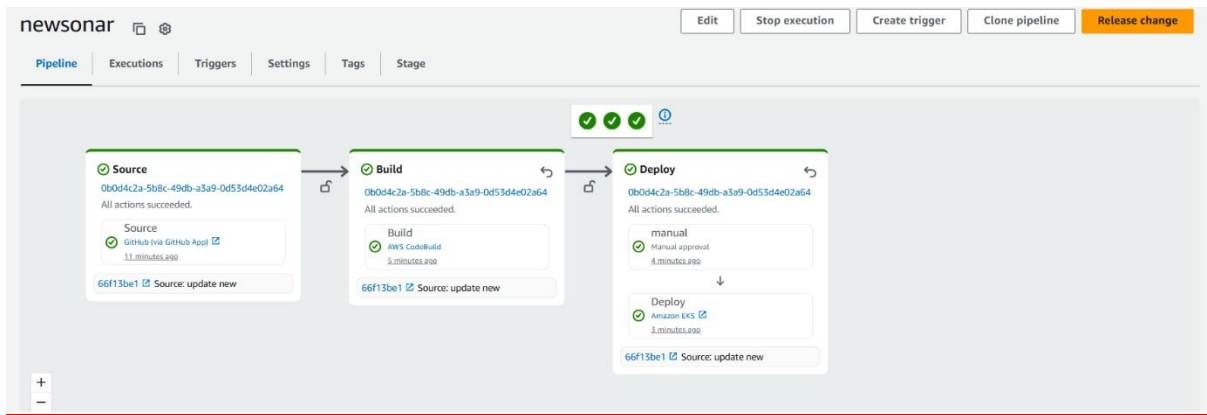
Decision

☐ Approve
Approving will resume the pipeline execution.

☐ Reject
Rejecting will stop the pipeline execution with a failed status.

Comments - optional ☐ Preview markdown [Learn more](#)

Cancel Submit



Web architecture

